



ScoutVLA: UAV-Centric Active Perception via a Dual-Expert VLA Model for Open-World Embodied Question Answering

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Abstract—Aerial Embodied Question Answering (EQA) requires Unmanned Aerial Vehicles (UAVs) to actively perceive the environment and answer natural language questions. Existing outdoor EQA systems usually stop once the target enters the UAV’s field of view, leaving the fine-grained viewpoint adjustment needed for evidence-seeking questions largely unresolved. To address this issue, we introduce FG-EQA, a fine-grained active perception EQA benchmark with more than 40K simulated trajectories and 1K real-world trajectories. Drawing inspiration from the “waggle dance” of scout bees, which iteratively adjust their flight paths to verify target information, we propose ScoutVLA, an evidence-driven Vision-Language-Action model for outdoor EQA. To emulate this active exploration behavior, ScoutVLA features a decoupled dual-expert architecture: a vision-language expert infers the semantic intent to identify missing evidence, while an independent action expert employs high-DoF flow matching to generate continuous viewpoint-refinement trajectories. To balance the competing demands of continuous control and semantic reasoning, we devise a decoupled training strategy with a knowledge insulation mechanism that prevents the action gradients from erasing the model’s multimodal reasoning ability. Extensive simulation experiments show that ScoutVLA-Pi0.5 achieves strict success and QA correctness rates that are $10.48\times$ and $7.85\times$ the corresponding baseline averages, respectively. Closed-loop real-world experiments further assess the deployability of the adapted models through quantitative results and inspection sequences. Our code, model checkpoints, dataset and supplementary materials are available at <https://anonymous.4open.science/w/ScoutVLA-B31F/>

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have emerged as flexible mobile sensors for urban monitoring and scene understanding. A compelling advancement in this domain is aerial Embodied Question Answering (EQA), where a UAV explores an environment to gather visual evidence for answering natural language questions. However, current outdoor EQA pipelines typically adopt a decoupled navigate-then-answer paradigm [1], [2], in which a UAV agent halts when the target enters the field of view or is in close proximity. Unfortunately, this premature termination leaves the UAV at poor viewing angles and distances, obscuring key visual evidence needed for accurate QA generation.

The inherent limitations of this paradigm are illustrated in Figure 1. In the initial view, a white truck is already clearly visible on the road. When tasked to read its license plate, standard baselines falter in varied ways: a passive pure vision-language model (e.g., Qwen3-VL) struggles due to the extreme distance, while a conventional EQA agent (e.g., CityEQA) might successfully fly closer but arbitrarily halt alongside the truck’s flank, completely missing the

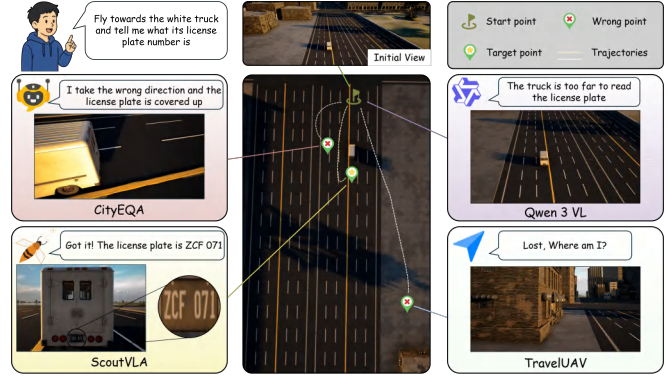


Fig. 1. Task illustration of question-conditioned fine-grained active perception.

rear-mounted plate. Moreover, the navigation model (e.g., TravelUAV) gets lost and completely deviates from the target direction. Acquiring precise, localized evidence requires the UAV to interpret the question’s semantic intent and execute continuous spatial maneuvers to align with specific locations like the rear bumper. Recognizing this critical deficit in current studies, we formalize this neglected late-stage EQA phase as *question-conditioned fine-grained active perception*.

Addressing this active perception task presents three formidable challenges. First, open-world visual evidence is exceptionally viewpoint-sensitive. A fractional deviation in spatial position and camera pose dictates whether a tiny, angle-dependent feature is legible. Second, establishing a semantic-to-physical mapping is inherently difficult. The agent must bridge a vast cross-modal gap, translating highly abstract linguistic intents directly into concrete 3D geometric search strategies [3], [4]. Third, fine-grained viewpoint control requires coordinated adjustments of UAV position, body yaw, and gimbal pitch. The policy must select the direction and magnitude of these adjustments to expose question-relevant details, which can be difficult to express through a fixed set of discrete actions.

To systematically address this problem, we introduce **FG-EQA**, the first benchmark dedicated to fine-grained perception in aerial EQA. Unlike long-horizon navigation datasets, each episode in FG-EQA isolates the evidence-seeking phase: the target is visible at the onset, yet necessitates a complex sequence of active viewpoint refinements to accurately answer the query. Specifically, it includes over 40K simulated trajectories and 1K human-piloted real-world

trajectories.

Furthermore, inspired by the “waggle dance” of scout bees, which iteratively hover and refine their flight paths to confirm discoveries, we propose **ScoutVLA**, a powerful dual-expert Vision-Language-Action (VLA) model trained with a decoupled strategy. To tackle the semantic-to-physical mapping, a PaliGemma-based multimodal expert perfectly aligns the question semantics with the missing visual evidence. To support this coordinated viewpoint control, an independent action expert uses flow matching to predict real-valued increments in UAV position, body yaw, and gimbal pitch. Finally, to prevent continuous action regression from catastrophically overriding discrete language reasoning, we devise a two-stage training paradigm. By integrating knowledge insulation with parameter-efficient LoRA and LMAdapter modules, we successfully orchestrate evidence-grounded QA without inducing catastrophic forgetting of continuous navigation capabilities.

The main contributions of this work are as follows:

- We formalize question-conditioned fine-grained active perception for aerial EQA and introduce FG-EQA, a benchmark comprising over 40K simulated trajectories and 1K human-piloted real-world trajectories.
- We propose ScoutVLA, a dual-expert VLA model for real-valued 5-DoF viewpoint control and terminal answer generation. A two-stage training strategy with knowledge insulation limits interference between action and language learning.
- Simulation experiments on FG-EQA show that ScoutVLA-Pi0.5 achieves strict success and QA correctness rates that are $10.48\times$ and $7.85\times$ the corresponding baseline averages, respectively. We further assess both adapted variants through closed-loop real-world inspection.

II. RELATED WORK

Embodied question answering. Early EQA focused on indoor navigation coupled with QA [5]–[9]. As studies transitioned to outdoor environments [1], [2], [10], they adopted modular navigate-then-answer pipelines. However, these systems implicitly assume that reaching a target’s general vicinity resolves the query, thereby entirely overlooking the fine-grained viewpoint adjustment necessary for angle-dependent visual evidence. ScoutVLA specifically targets this missing active perception phase.

Active perception and aerial VLA models. Recent multi-modal autonomous agents [11]–[16] and aerial VLA models [17]–[22] have successfully mapped language instructions to flight control. Yet, despite their active mobility, they primarily execute macroscopic routing (e.g., “fly to the building”) utilizing low-DoF or discrete action spaces. ScoutVLA advances this paradigm by integrating the linguistic query directly into continuous 5-DoF control to actively reveal occluded evidence. A comprehensive discussion of related literature, alongside an architectural comparison table of recent VLA models, is provided in the supplementary materials.

III. FG-EQA BENCHMARK

A. Task Formulation

Each episode starts from an initial UAV view in which the target is visible, but the available visual cues are insufficient to correctly answer the question. At time t , the UAV receives an RGB perception I_t , state s_t , and question q , executes a viewpoint-refinement command $\mathbf{u}_t$, and finally produces a short answer y . The goal is to acquire visual evidence sufficient to answer the question. We express this task-level objective in Equation 1 as maximizing the expected log-likelihood of the ground-truth short answer y^* over trajectories τ induced by policy π_θ . Multiple trajectories may provide sufficient evidence for the same question; therefore, the task does not assume a unique reference path.

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\tau \sim \pi_\theta} [\log p_\theta(y^* | \tau, q)]. \quad (1)$$

For fine-grained perception, the UAV acts in a 5-DoF viewpoint-refinement space, $\mathbf{u}_t = [\Delta x_t, \Delta y_t, \Delta z_t, \Delta \psi_t, \Delta \theta_t^g]$, where the first three dimensions translate the UAV and the last two adjust body yaw and gimbal pitch. State and model-prediction action encodings are provided in the supplementary materials.

B. Dataset Construction

To bridge the gap between macroscopic spatial reasoning and fine-grained visual evidence acquisition, we construct the FG-EQA dataset through a systematic two-phase pipeline: (1) Perception Trajectory Generation, which formulates viewpoint-refinement flight paths; and (2) QA Data Annotation, which aligns terminal evidence with complex natural language queries.

Perception Trajectory Generation. We initiate episodes by reusing simulated outdoor scenes from UAV-ON [23]. To construct varied starting poses, we apply yaw offsets ($\pm 15^\circ$, $\pm 30^\circ$) from the target-oriented optical axis. Three vision-language models (VLMs), Qwen3-VL-235B-A22B-Instruct, Kimi-K2.6, and GLM-5.1, classify each initial view as fully visible, partially visible, or fully invisible. We accept unanimous labels and use human adjudication for disagreements, retaining only fully or partially visible targets. We uniformly sample candidate terminal viewpoints on a target-centered upper hemisphere. UAV experts retain views in which the target is clear, unoccluded, and complete. For each selected start–endpoint pair, the base trajectory follows the straight-line segment connecting the two positions. The initial view retains the yaw offset. Once translation begins, the UAV body yaw is adjusted to face the target and continuously updated during motion. Trajectories with detected collisions are discarded. To endow ScoutVLA with dynamic error-correction capabilities, we inject random pose perturbations into 30% of the trajectories to act as negative recovery samples. Final manual replay removes trajectories with visual defects or simulation anomalies. To support real-world adaptation, we also collect 1K human-piloted trajectories. These physical episodes are captured by professional UAV pilots manually executing analogous micro-viewpoint refinement maneuvers

around real-world targets, recording unconstrained onboard camera streams from initial discovery to final evidence acquisition.

QA Data Annotation. Based on the validated endpoint views, we define seven critical outdoor question types. The three VLMs each generate one third of the candidate questions from terminal views before filtering. Each question is randomly assigned to one model from the same pool to generate a short answer and reasoning chain. A triplet passes terminal-view verification only if all three models approve it. Gemini 3 Flash and GPT-5.1 then answer using only the initial image and question. Given the question, reference answer, and each response, DeepSeek-V4-Pro returns a binary correctness judgment; we discard the sample if either response is judged correct. Human annotators audit a subset of the remaining QA samples. We additionally incorporate 80K generic VQA samples from LLaVA-Instruct-150K [24] to broaden the visual and semantic coverage of language training. Model versions, annotation and filtering procedures, and dataset statistics are provided in the supplementary materials.

IV. SCOUTVLA

ScoutVLA integrates multimodal reasoning and continuous viewpoint control in an end-to-end dual-expert architecture. This model consists of two key modules: a PaliGemma 2-based multimodal expert [25], and an independent action expert network. Furthermore, we introduce a decoupled two-stage training strategy to ensure that the UAV agent establishes robust flight control capabilities while successfully retaining visual question-answering skills. The comprehensive system overview is illustrated in Figure 2.

A. Architecture

ScoutVLA-Pi0 and ScoutVLA-Pi0.5 are initialized from the pretrained π_0 and $\pi_{0.5}$ checkpoints, respectively. Both variants adapt the pretrained vision-language and action experts to UAV active perception through architectural modifications and the two-stage post-training procedure described below.

The vision-language expert is built on PaliGemma 2 and handles visual encoding and semantic reasoning. Visual tokens extracted by SigLIP [26] are contextually fused with language inputs through bidirectional attention. The fused visual-language representation provides question-conditioned context for action prediction.

The second core component is an independent action expert for high-DoF trajectory generation. Conditioned on this context and the UAV state s_t , the action expert predicts continuous 5-DoF action chunks through flow matching [27]. As new observations become available, the policy updates its action predictions to form a closed-loop viewpoint-adjustment trajectory. This design bypasses discrete action quantization, directly targeting the fine-grained spatial maneuvers required to overcome viewpoint sensitivity.

For terminal answer generation, we insert LMAdapter between the multimodal backbone and the language output

head to adapt the hidden representations used for token prediction. LoRA provides low-rank updates within the LLM backbone, while LMAdapter is a separate module at its output; both are trained during Stage 2 language updates. After viewpoint control terminates, the language branch generates a short answer from the terminal image and the original question.

B. Two-Stage Training Recipe

Action-loss updates to a shared pretrained backbone can cause catastrophic forgetting of its language capabilities [28], [29]. To limit this interference, ScoutVLA first learns viewpoint control in Stage 1 and then applies task-specific gradient routing during action-language training in Stage 2.

Stage 1: action-only control learning. Given a clean action chunk $\mathbf{u}$ sampled from the retained demonstration trajectories, Gaussian noise ϵ , interpolation scalar λ , interpolated sample $\mathbf{x}_\lambda = \lambda\epsilon + (1-\lambda)\mathbf{u}$, and multimodal condition $\mathbf{c}$, Stage 1 optimizes a continuous vector field objective to minimize the action denoising error, as shown in Equation 2. Here, $\mathbf{c}$ comprises the visual-language context and UAV state information supplied to the action expert. During this stage, all trainable model parameters are updated to establish the visual-action mapping.

$$\mathcal{L}_{\text{stage1}} = \mathbb{E}_{\lambda, \mathbf{u}, \epsilon} \left[\|\mathbf{v}_\theta(\mathbf{x}_\lambda, \lambda, \mathbf{c}) - (\epsilon - \mathbf{u})\|_2^2 \right]. \quad (2)$$

Stage 2: action-language joint training with knowledge insulation. Stage 2 starts from the Stage 1 checkpoint and alternates action and language updates. The base LLM weights remain frozen throughout this stage. During action updates, gradients are blocked from propagating into the LLM backbone, including its LoRA modules, and only the action expert is optimized. During language updates, the backbone is adapted through LoRA and LMAdapter is fully fine-tuned, while the action expert remains fixed. The per-step objective is defined in (3).

$$\mathcal{L}_{\text{stage2}} = \begin{cases} \mathcal{L}_{\text{action}}, & \text{action step,} \\ \mathcal{L}_{\text{language}}, & \text{language step.} \end{cases} \quad (3)$$

Here, $\mathcal{L}_{\text{action}}$ is the flow-matching loss in (2). Given the terminal FPV image I_T and question q , $\mathcal{L}_{\text{language}}$ is the short-answer next-token loss defined in (4). w_j is the weight.

$$\mathcal{L}_{\text{language}} = - \sum_{j=1}^{|y|-1} w_j \log p_\theta(y_{j+1} \mid I_T, q, y_{1:j}). \quad (4)$$

V. EXPERIMENTS

To evaluate ScoutVLA, we design comprehensive simulated and real-world experiments to answer four research questions: **(RQ1)** Does ScoutVLA outperform the state-of-the-art baselines on the active perception task? **(RQ2)** How do the two-stage training paradigm and specific language-recovery modules contribute? **(RQ3)** Does the active perception process yield more answerable evidence at final

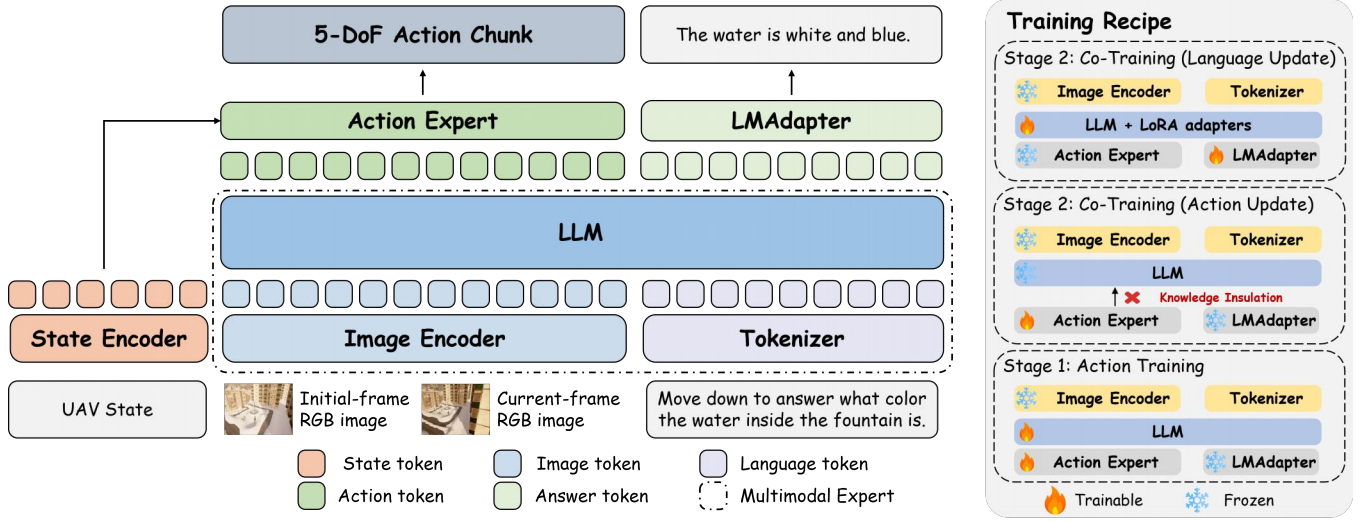


Fig. 2. ScoutVLA overview. Left: dual-expert architecture for question-conditioned viewpoint control and short-answer generation. Right: two-stage training recipe.

viewpoint? (**RQ4**) How do the adapted ScoutVLA models perform in closed-loop real-world inspection?

A. Setup

Environments and Metrics. All simulation experiments use nine environments shared between training and testing. For each target instance in each environment, we reserve 10–20 trajectories for evaluation, with initial positions and terminal viewpoints different from those used for training. Test trajectories, terminal images, and questions are excluded from training. This protocol evaluates fine-grained active perception on held-out trajectories and questions around previously observed targets. We evaluate navigation quality using Target Detection Rate (TDR), terminal Distance to Target (DT), and Average Trajectory Length (ATL). Answer quality is measured by LLM-judged QA Score (QAS) and QA Correctness (QAC). Strict Success Rate (SSR) measures end-to-end task success. Success is assessed by target visibility and answer correctness at the final view, without requiring the executed trajectory to match a reference path. Following prior work on open-ended EQA [6], we use LLM-based semantic evaluation to accommodate valid answer paraphrases. Qwen3-VL-235B-A22B, Kimi-K2.6, and GLM-5.1 each score interpretable generated answers on a 1–5 scale given the question and reference answer. The same scoring rubric is applied to all methods. We average the three scores and consider an answer correct when the mean is at least 4. For all methods, uninterpretable outputs, including garbled text, are assigned the minimum score of 1 and counted as incorrect. Full evaluation details are provided in the supplementary materials.

Implementation Details. Table I summarizes the two-stage training configuration for both ScoutVLA variants. Stage 1 uses AdamW with cosine learning-rate decay, a 1,000-step linear warmup, and an exponential moving average (EMA) decay of 0.99. Stage 2 uses three action batches per language batch and a $2.5\times$ learning-rate multiplier for

TABLE I
TWO-STAGE TRAINING CONFIGURATION.

Parameter	Value
Batch Size	128
Epochs per Stage	5
Peak Learning Rate	2×10^{-5}
LoRA Enabled	True (Stage 2)
LoRA Rank	16
Action Horizon Steps	10
Training Time	~ 120 hours per stage
Hardware	$8 \times$ NVIDIA A800 GPUs

LMAdapter. All Qwen3-VL models used in the experiments are Instruct variants and the suffix is omitted for brevity.

B. Baselines

We compare ScoutVLA with baselines covering seven evaluation families. **Random** executes randomly sampled navigation actions before final-frame QA. **Passive QA** includes blind text-only LLM baselines, which receive the question without visual input to test language priors, and an initial-view VQA baseline, which answers directly from the start frame. **General VLM** evaluate whether general multimodal models can act as zero-shot perception agents. **Navigation** baselines, including OpenFly [17] and TravelUAV [19], are designed for UAV navigation rather than answer generation. **Aerial VLA** uses UAV-Flow [21], which generates UAV actions but has no native QA head, to test whether continuous UAV action generation alone is sufficient. Policy-only baselines without native answer generation are evaluated by using Qwen3-VL-235B-A22B to answer from their terminal FPV image and the original question. **Indoor EQA** and **Outdoor EQA** are represented by CoV [30] and CityEQA [1], respectively.

C. Comparison with State of the Art

To answer RQ1, we evaluate end-to-end active perception performance. Results are summarized in Table II. As shown, Passive QA baselines obtain low QAS and QAC, indicating that language priors or the initial view alone rarely contain sufficient answer evidence. General VLM and policy-only baselines can sometimes localize the target, as reflected by the 57.41% TDR of Qwen2-VL-2B, but their terminal viewpoints remain insufficient for reliable fine-grained EQA. Existing indoor and outdoor EQA methods improve over generic navigation policies, yet the best non-ScoutVLA baseline reaches only 11.79% SSR and 16.96% QAC. In contrast, ScoutVLA-Pi0.5 achieves 71.90% SSR, 80.99% TDR, and 72.70% QAC, while ScoutVLA-Pi0 obtains the lowest terminal distance. Question-conditioned continuous perception significantly improves both missing evidence acquisition and final EQA success over all existing paradigms.

D. Ablation on Training Stages and Components

To address RQ2, Table III compares training stages and Stage 2 components. Stage 1 rows report viewpoint-control performance, while Stage 2 rows correspond to the full models in Table II. ScoutVLA-Pi0.5 reaches 71.90% SSR, 72.70% QAC, and 80.99% TDR after Stage 2.

All component ablations start from the same ScoutVLA-Pi0.5 Stage 1 checkpoint, with other training settings unchanged. Without LoRA, language updates train only LMAdapter. Without LMAdapter, the original language output head is retained and language updates train only LoRA. Without knowledge insulation, action-loss gradients also update the shared LoRA parameters, providing a joint-training comparison in which both tasks update these parameters. The base LLM weights remain frozen in all Stage 2 variants.

Removing LoRA changes TDR from 80.99% to 80.85%, while QAC falls from 72.70% to 12.32%. Removing LMAdapter yields 80.52% TDR, but all answers are garbled, yielding 0% QAC and SSR under the invalid-output rule. Removing knowledge insulation reduces TDR to 52.12% and QAC to 10.32%. Under this configuration, the LoRA and LMAdapter ablations primarily affect answering, while removing knowledge insulation degrades both target detection and answering.

E. Terminal-View Answerability and VQA Quality

To answer RQ3, we decouple the evaluation of the generated terminal views from the inherent language-side answering capability. Table IV first evaluates terminal-view answerability. We use Qwen3-VL-8B, Qwen3-VL-32B, and Qwen3-VL-235B-A22B as a shared panel of external VQA answerers. For each view source, we hold the terminal FPV image fixed, ask the same external VQA model panel to answer the original question, and score the generated answers with the standard judge protocol. As shown, ScoutVLA-Pi0.5 achieves the highest mean QAS and QAC, indicating that its policy produces terminal views with significantly more complete and discriminative visual evidence compared to other baselines.

Furthermore, Table V isolates language-side VQA ability on held-out UAV-scene VQA samples. Qwen2-VL-2B is included as a comparable-scale general VLM baseline, while Qwen3-VL-235B-A22B serves as a strong general-purpose reference. ScoutVLA-Pi0.5 reaches 4.78 QAS and 95.40% QAC, outperforming both baselines. These results show higher QA performance on the held-out FG-EQA samples. The terminal-view analysis also shows improved answerability for the evaluated VQA models.

F. Human Validation of QA Evaluation

Following the validation design of OpenEQA [6], we assess agreement between automated QA evaluation and human judgments on 300 held-out question-response instances covering 6 evaluated methods and seven question categories. 4 human evaluators independently assign 1–5 scores and binary correctness labels, without access to response sources or automated judgments. We compare the mean human score with the unrounded mean score from our three LLM judges. We observe a Spearman correlation of $\rho = 0.84$ (95% CI: [0.80, 0.87]), with a mean human-human correlation of 0.88. Human and LLM scores are highly correlated on this subset, supporting the consistency of automated QA scoring with human assessment. Annotation procedures and correctness-agreement results are provided in the supplementary materials.

G. Real-World Experiments

Experimental Setup. To address RQ4, we evaluate closed-loop inspection on an Amov Lab P600 UAV (Figure 3) equipped with a gimbal-mounted monocular RGB camera, a real-time kinematic (RTK) module, and GPS. ScoutVLA runs on a ground station equipped with an NVIDIA RTX 5090 GPU. FPV images are transmitted wirelessly to the remote controller (RC) and forwarded to the ground station through a wired connection. Given the observations and question, ScoutVLA generates continuous 5-DoF viewpoint commands, which return through the RC to the flight controller.

Real-world data collection and experiments were conducted with authorization from the relevant site and vehicle owners and consent from involved individuals. Faces and other sensitive visual information were pixelated. Flights were registered with the local authorities and conducted under human-pilot supervision, with flight-area restrictions and loss-of-link procedures.

Runtime Performance. Table VI reports action inference latency, image transfer latency, and action execution rate for both ScoutVLA variants. Action inference latency measures the time to generate a complete 10-step action chunk. Image transfer latency spans onboard image acquisition to decoded-frame availability at the ground station. The UAV executes all ten actions sequentially before the next inference pass.

Both ScoutVLA variants are initialized from checkpoints trained on the 40K simulated trajectories. We first fine-tune the action expert on 1,000 human-piloted perception

TABLE II
END-TO-END ACTIVE PERCEPTION PERFORMANCE ON FG-EQA.

Type	Method	SSR (%) $\uparrow$	TDR (%) $\uparrow$	DT (m) $\downarrow$	ATL (m)	QAS $\uparrow$	QAC (%) $\uparrow$
Random	Random	9.86 ± 0.57	47.57 ± 4.19	33.80 ± 8.53	23.92 ± 12.41	1.70 ± 0.07	13.58 ± 1.18
Passive QA	Blind (V4 Pro) [31]	–	–	–	–	1.26 ± 0.11	6.60 ± 1.34
	Blind (V4 Flash) [31]	–	–	–	–	1.13 ± 0.08	3.30 ± 0.82
	Initial (Qwen3-VL-235B-A22B) [32]	–	–	–	–	1.50 ± 0.16	9.70 ± 2.13
General VLM	Qwen2-VL-2B [33]	10.35 ± 1.87	57.41 ± 3.82	40.60 ± 9.21	6.10 ± 2.19	1.60 ± 0.19	13.00 ± 2.48
	Qwen3-VL-8B [32]	5.73 ± 1.23	37.04 ± 4.47	42.90 ± 10.42	45.30 ± 14.86	1.36 ± 0.13	6.83 ± 1.46
	Qwen3-VL-235B-A22B [32]	2.86 ± 0.98	22.22 ± 3.11	76.35 ± 16.54	102.20 ± 28.63	1.30 ± 0.14	6.39 ± 1.28
Navigation	OpenFly [17]	4.63 ± 1.09	19.91 ± 2.87	65.50 ± 14.18	72.20 ± 18.37	1.48 ± 0.17	10.57 ± 2.14
	TravelUAV [19]	0.00 ± 0.00	12.96 ± 2.41	<u>26.60 ± 7.83</u>	46.00 ± 12.52	1.17 ± 0.11	3.08 ± 0.77
Aerial VLA	UAV-Flow [21]	5.73 ± 1.38	29.17 ± 3.63	79.70 ± 15.82	93.40 ± 24.54	1.49 ± 0.18	9.03 ± 1.81
Indoor EQA	CoV [30]	10.79 ± 2.16	36.11 ± 4.23	52.80 ± 11.38	44.90 ± 13.24	1.80 ± 0.21	16.96 ± 2.78
Outdoor EQA	CityEQA [1]	<u>11.79 ± 1.97</u>	39.20 ± 3.92	35.70 ± 8.57	10.90 ± 3.44	1.43 ± 0.14	<u>12.11 ± 2.26</u>
Ours	ScoutVLA-Pi0	60.78 ± 3.48	76.10 ± 2.89	16.20 ± 3.16	58.95 ± 7.81	3.45 ± 0.24	62.20 ± 3.58
	ScoutVLA-Pi0.5	71.90 ± 2.83	80.99 ± 2.18	17.56 ± 2.76	62.04 ± 6.54	3.91 ± 0.19	72.70 ± 2.92

Values are mean $\pm$ standard deviation. Arrows indicate preferred directions. ATL measures path length and should be interpreted together with SSR and TDR. “–” indicates that the metric is not applicable because the method does not perform navigation. Bold values denote the best overall results, and underlined values denote the best non-ScoutVLA baselines.

TABLE III
ABLATION ON TRAINING STAGES AND STAGE 2 COMPONENTS.

Model	SSR (%) $\uparrow$	TDR (%) $\uparrow$	DT (m) $\downarrow$	ATL (m)	QAS $\uparrow$	QAC (%) $\uparrow$
<i>Effect of two-stage training</i>						
ScoutVLA-Pi0-Stage1	–	71.91 ± 2.95	18.06 ± 3.21	60.72 ± 7.54	–	–
ScoutVLA-Pi0-Stage2	60.78 ± 3.48	76.10 ± 2.89	16.20 ± 3.16	58.95 ± 7.81	3.45 ± 0.24	62.20 ± 3.58
ScoutVLA-Pi0.5-Stage1	–	76.04 ± 2.51	18.01 ± 2.94	62.07 ± 6.82	–	–
ScoutVLA-Pi0.5-Stage2	71.90 ± 2.83	80.99 ± 2.18	17.56 ± 2.76	62.04 ± 6.54	3.91 ± 0.19	72.70 ± 2.92
<i>Ablation on Stage 2 components based on ScoutVLA-Pi0.5</i>						
w/o LoRA	10.28 ± 3.31	80.85 ± 2.15	17.61 ± 2.79	62.11 ± 6.48	2.48 ± 0.26	12.32 ± 1.47
w/o knowledge insulation	8.04 ± 3.52	52.12 ± 3.42	22.43 ± 4.51	58.12 ± 8.17	1.48 ± 0.26	10.32 ± 1.47
w/o LMAdapter	0.00 ± 0.00	80.52 ± 2.21	17.65 ± 2.84	61.98 ± 6.63	1.00 ± 0.00	0.00 ± 0.00

-- denotes metrics not reported for Stage 1 checkpoints.



Fig. 3. Amov Lab P600 UAV platform used for real-world inspection.

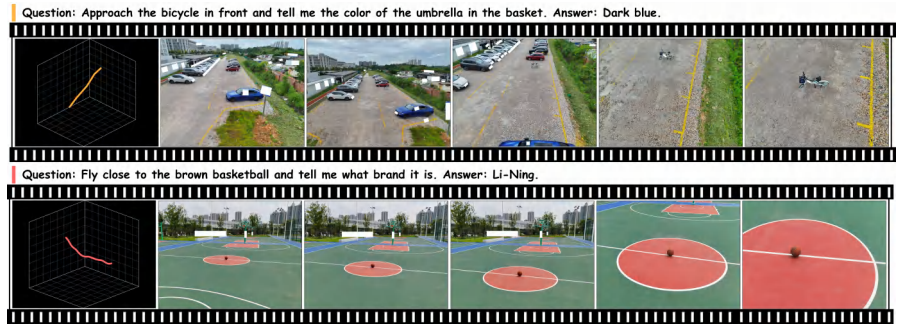


Fig. 4. Real-world inspection examples.

trajectories, then adapt the language components with real-image VQA data, excluding AirSim-generated VQA pairs during adaptation. The demonstration set contains 200 trajectories per target category and is distinct from the closed-loop evaluation flights. We compare the adapted variants with

DeepSeek-4.1-Flash, GLM-5.3-Flash, and Qwen3-VL-235B-A22B. Other baselines are not included due to incompatibility with our UAV platform

The five evaluation tasks ask for the brand of a basketball, the color of an umbrella in a bicycle basket, whether anyone

TABLE IV

TERMINAL-VIEW ANSWERABILITY UNDER A SHARED VQA PANEL.

View source	Metric	Qwen3-VL-8B	Qwen3-VL-32B	Qwen3-VL-235B-A22B	Mean
Qwen2-VL-2B	QAS	1.42	1.45	1.50	1.46
	QAC (%)	8.80	7.70	9.70	8.73
Qwen3-VL-8B	QAS	1.36	1.42	1.36	1.38
	QAC (%)	6.83	7.70	6.80	7.11
Qwen3-VL-235B-A22B	QAS	1.11	1.17	1.30	1.19
	QAC (%)	2.20	2.00	6.39	3.53
ScoutVLA-Pi0	QAS	2.02	2.36	2.75	2.38
	QAC (%)	25.60	33.90	43.80	34.43
ScoutVLA-Pi0.5	QAS	2.16	2.56	2.82	2.51
	QAC (%)	28.90	34.30	45.50	36.23

TABLE V

LANGUAGE-SIDE VQA QUALITY.

Model	QAS	QAC (%)
Qwen2-VL-2B	3.21	50.40
Qwen3-VL-235B-A22B	4.13	77.50
ScoutVLA-Pi0	4.67	92.10
ScoutVLA-Pi0.5	4.78	95.40

is inside a black car, whether anyone is inside a white truck, and the sign on a signboard. Each model completes 10 trials per task, yielding 50 flights per model and 250 flights overall. Each episode starts with a visible target whose question-relevant details require viewpoint refinement.

Quantitative Results. In Table VII, Qwen3-VL-235B-A22B is the strongest baseline with 40.00% SSR. ScoutVLA-Pi0 and ScoutVLA-Pi0.5 achieve 70.00% and 72.00% SSR, improving by 30 and 32 percentage points, respectively, while also obtaining higher TDR and QAS and lower DT. Relative to ScoutVLA-Pi0, ScoutVLA-Pi0.5 improves TDR from 76.00% to 82.00% and QAS from 4.06 to 4.20, with a slightly lower DT. Its 2-point SSR gain corresponds to one additional successful flight, alongside a longer ATL.

Qualitative Cases. Figure 4 shows two real-world inspection examples, each with a flight trajectory and five selected FPV frames. In the bicycle example, the UAV approaches the basket to inspect an umbrella and answers that its color is dark blue. In the basketball example, the UAV approaches the ball on the court and identifies its brand as Li-Ning. These examples illustrate how viewpoint refinement provides closer observations of question-relevant details. Additional simulated success cases and real-world failure cases are analyzed in the supplementary materials.

VI. LIMITATIONS

ScoutVLA has several limitations. First, the policy assumes that the target is already visible and focuses on viewpoint refinement after target discovery. Second, our simulated base demonstrations use straight-line connections, and trajectories with detected collisions are discarded. This limits coverage of obstacle-avoidance maneuvers, and performance in tasks requiring detours remains to be systematically evaluated. Expert screening favors clear, unobstructed terminal views and may underrepresent tasks with persistent occlusion or difficult access to informative viewpoints. Third, trajectory smoothness is not quantitatively evaluated. Continuous action

TABLE VI

RUNTIME PERFORMANCE IN REAL-WORLD DEPLOYMENT.

Metric	ScoutVLA-Pi0	ScoutVLA-Pi0.5
Action inference latency (ms)	47.3	52.9
Image transfer latency (s)	2.614	2.702
Action execution rate (Hz)	1.974	1.795

prediction alone does not establish smooth executed motion. Future evaluation should assess translational acceleration and jerk, as well as body-yaw and gimbal-pitch dynamics. Finally, the benchmark mainly covers static targets and single-agent operation, while real-world testing includes a limited set of targets and flights. Broader physical benchmarking is needed across sites and conditions.

VII. CONCLUSION

In aerial EQA, existing paradigms prematurely halt upon macroscopic target localization, leaving the fine-grained viewpoint adjustment required to gather hidden visual evidence largely unresolved. To address this gap, we introduce FG-EQA and ScoutVLA, which couples a multimodal expert for semantic intent inference with a continuous flow-matching expert for viewpoint control. Our decoupled two-stage training strategy supports continuous control and textual answering through knowledge insulation. Simulation experiments demonstrate improvements in active perception and EQA, and closed-loop real-world experiments assess both adapted variants through quantitative metrics and inspection sequences. Together, these experiments examine how question-conditioned viewpoint refinement supports evidence acquisition across simulated and physical settings. Future work will integrate this policy with long-horizon search agents and extend evaluation to dynamic targets and broader real-world conditions.

REFERENCES

- [1] Y. Zhao, K. Xu, Z. Zhu, Y. Hu, Z. Zheng, Y. Chen, Y. Ji, C. Gao, Y. Li, and J. Huang, "CityEQA: A hierarchical LLM agent on embodied question answering benchmark in city space," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 12465–12480.
- [2] S. Varghese, J. Gao, A. U. Rahman, and V. Hoskere, "BridgeEQA: Virtual embodied agents for real bridge inspections," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 8163–8173.
- [3] S. Wu, R. Liu, Z. Xie, and Z. Pang, "Map-SemNav: Advancing zero-shot continuous vision-and-language navigation through visual semantics and map integration," in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2025, pp. 7577–7584.
- [4] P. Kong, R. Liu, Z. Xie, and Z. Pang, "VLN-KHVR: Knowledge-and-history aware visual representation for continuous vision-and-language navigation," in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2025, pp. 5236–5243.
- [5] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, "Embodied question answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 1–10.
- [6] A. Majumdar, A. Ajay, X. Zhang, P. Putta, S. Yenamandra, M. Henaff, S. Silwal, P. McVay, O. Maksymets, S. Arnaud, et al., "OpenEQA: Embodied question answering in the era of foundation models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16488–16498.

TABLE VII
REAL-WORLD CLOSED-LOOP INSPECTION PERFORMANCE.

Method	TDR (%) $\uparrow$	DT (m) $\downarrow$	ATL (m)	QAS $\uparrow$	QAC (%) $\uparrow$	SSR (%) $\uparrow$
DeepSeek-4.1-Flash	16.00	16.42	40.90	1.62	14.00	14.00
GLM-5.3-Flash	6.00	19.45	36.52	1.30	6.00	6.00
Qwen3-VL-235B-A22B	40.00	13.70	34.14	2.62	40.00	40.00
ScoutVLA-Pi0	76.00	6.10	23.16	4.06	70.00	70.00
ScoutVLA-Pi0.5	82.00	5.93	25.38	4.20	72.00	72.00

- [7] K. Cheng, Z. Li, X. Sun, B.-C. Min, A. S. Bedi, and A. Bera, "EfficientEQA: An efficient approach to open-vocabulary embodied question answering," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025, pp. 20357–20364.
- [8] M. Zhai, Z. Gao, Y. Wu, and Y. Jia, "Memory-centric embodied question answering," *arXiv preprint arXiv:2505.13948*, 2025.
- [9] K. Jiang, Y. Liu, W. Chen, J. Luo, Z. Chen, L. Pan, G. Li, and L. Lin, "Beyond the destination: A novel benchmark for exploration-aware embodied question answering," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 9091–9101.
- [10] Y. Li, Y. Chen, A. Dao, L. Li, Z. Cai, Z. Tan, T. Chen, and Y. Kong, "IndustryEQA: Pushing the frontiers of embodied question answering in industrial scenarios," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [11] Y. Hu, A. Xi, Q. Xiao, S. Isaacson, H. X. Liu, R. Vasudevan, and M. Ghaffari, "LongNav-R1: Horizon-adaptive multi-turn RL for long-horizon VLA navigation," in *Proceedings of Robotics: Science and Systems*, 2026.
- [12] J. Zha, Y. Fan, T. Zhang, G. Chen, Y. Chen, C. Gao, and X. Chen, "AirCopBench: A benchmark for multi-drone collaborative embodied perception and reasoning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 2, 2026, pp. 1507–1515.
- [13] J. He, J. Fang, F. Xiong, Z. Yao, F. Shen, H. Guo, J. Wang, and T.-S. Chua, "Active Zero: Self-evolving vision-language models through active environment exploration," *arXiv preprint arXiv:2602.11241*, 2026.
- [14] P. Zhang, Z. Huang, Y. Wang, J. Zhang, L. Xue, Z. Wang, Q. Wang, K. Chandrasegaran, R. Zhang, Y. Choi, et al., "Theory of space: Can foundation models construct spatial beliefs through active exploration?" in *Proceedings of the International Conference on Learning Representations*, 2026.
- [15] S. Shah, D. Dey, C. Lovett, and A. Kapoor, "AirSim: High-fidelity visual and physical simulation for autonomous vehicles," in *Field and Service Robotics*. Cham, Switzerland: Springer, 2018, pp. 621–635.
- [16] Y. Ji, Z. Zhu, Y. Zhao, B. Liu, C. Gao, Y. Zhao, S. Qiu, Y. Hu, and Q. Yin, "Towards autonomous UAV visual object search in city space: Benchmark and agentic methodology," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 22, 2026, pp. 18342–18350.
- [17] Y. Gao, C. Li, Z. You, J. Liu, L. Zhen, P. Chen, Q. Chen, Z. Tang, L. Wang, Y. Tang, et al., "OpenFly: A comprehensive platform for aerial vision-language navigation," in *Proceedings of the International Conference on Learning Representations*, 2026.
- [18] H. Cai, J. Dong, J. Tan, J. Deng, S. Li, Z. Gao, H. Wang, Z. Su, A. Sumalee, and R. Zhong, "FlightGPT: Towards generalizable and interpretable UAV vision-and-language navigation with vision-language models," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 6659–6676.
- [19] X. Wang, D. Yang, Z. Wang, H. Kwan, J. Chen, H. Li, Y. Liao, S. Liu, et al., "Towards realistic UAV vision-language navigation: Platform, benchmark, and methodology," in *Proceedings of the International Conference on Learning Representations*, 2025.
- [20] T. Li, T. Huai, Z. Li, Y. Gao, H. Li, and X. Zheng, "SkyVLN: Vision-and-language navigation and NMPC control for UAVs in urban environments," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025, pp. 17199–17206.
- [21] X. Wang, D. Yang, Y. Liao, W. Zheng, B. Dai, H. Li, S. Liu, et al., "UAV-Flow Colosseo: A real-world benchmark for flying-on-a-word UAV imitation learning," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [22] Y. Wu, M. Zhu, X. Li, Y. Du, Y. Fan, W. Li, Z. Han, X. Zhou, and F. Gao, "VLA-AN: An efficient and onboard vision-language-action framework for aerial navigation in complex environments," *arXiv preprint arXiv:2512.15258*, 2025.
- [23] J. Xiao, Y. Sun, Y. Shao, B. Gan, R. Liu, Y. Wu, W. Guan, and X. Deng, "UAV-ON: A benchmark for open-world object goal navigation with aerial agents," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 13023–13029.
- [24] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 34892–34916.
- [25] A. Steiner, A. S. Pinto, M. Tschannen, D. Keysers, X. Wang, Y. Bitton, A. Gritsenko, M. Minderer, A. Sherbondy, S. Long, et al., "PaliGemma 2: A family of versatile VLMs for transfer," *arXiv preprint arXiv:2412.03555*, 2024.
- [26] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 11975–11986.
- [27] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *Proceedings of the International Conference on Learning Representations*, 2023.
- [28] Z. Zhou, Y. Zhu, M. Zhu, J. Wen, N. Liu, Z. Xu, W. Meng, Y. Peng, C. Shen, F. Feng, et al., "ChatVLA: Unified multimodal understanding and robot control with vision-language-action model," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 5377–5395.
- [29] Z. Zhou, Y. Zhu, X. Liu, Z. Tang, J. Wen, Y. Peng, C. Shen, and Y. Xu, "ChatVLA-2: Vision-language-action model with open-world reasoning," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 45537–45559.
- [30] H. Zhao, A. Liu, Z. Zhang, W. Wang, F. Chen, R. Zhu, G. Haffari, and B. Zhuang, "CoV: Chain-of-view prompting for spatial reasoning," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 32427–32440.
- [31] A. Xu, B. Lin, B. Xue, B. Wang, B. Xu, B. Wu, B. Zhang, C. Lin, C. Dong, C. Ling, et al., "DeepSeek-V4: Towards highly efficient million-token context intelligence," *arXiv preprint arXiv:2606.19348*, 2026.
- [32] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge, et al., "Qwen3-VL technical report," *arXiv preprint arXiv:2511.21631*, 2025.
- [33] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, et al., "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.